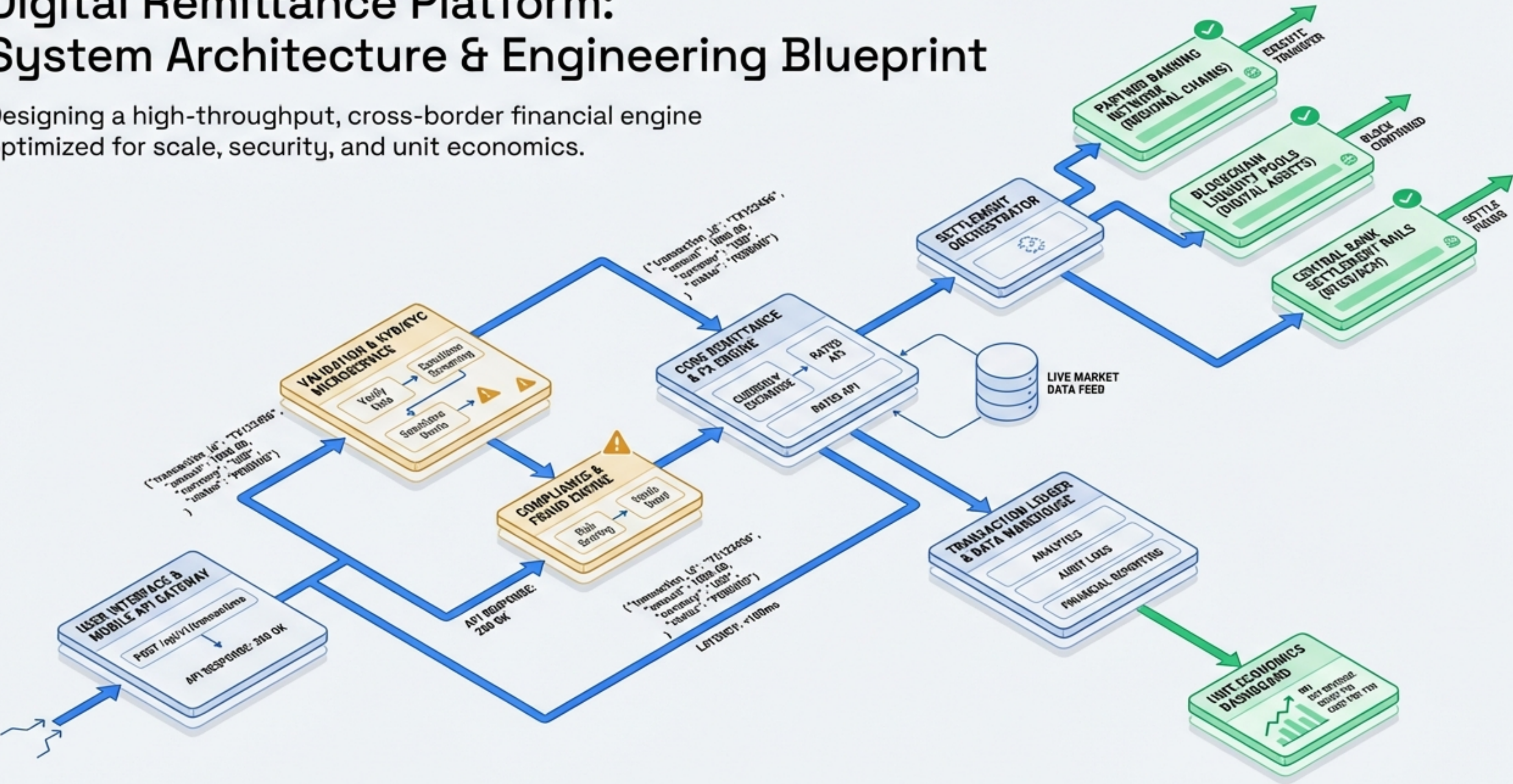


Digital Remittance Platform: System Architecture & Engineering Blueprint

Designing a high-throughput, cross-border financial engine optimized for scale, security, and unit economics.



The Engineering Challenge at Scale

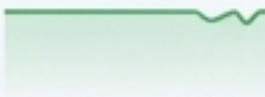
Throughput & Reach

- Daily Volume:
1M+
transfers/day 
- Active Corridors:
50+
global currency pairs 
- Concurrent Users:
100K+
at peak load 

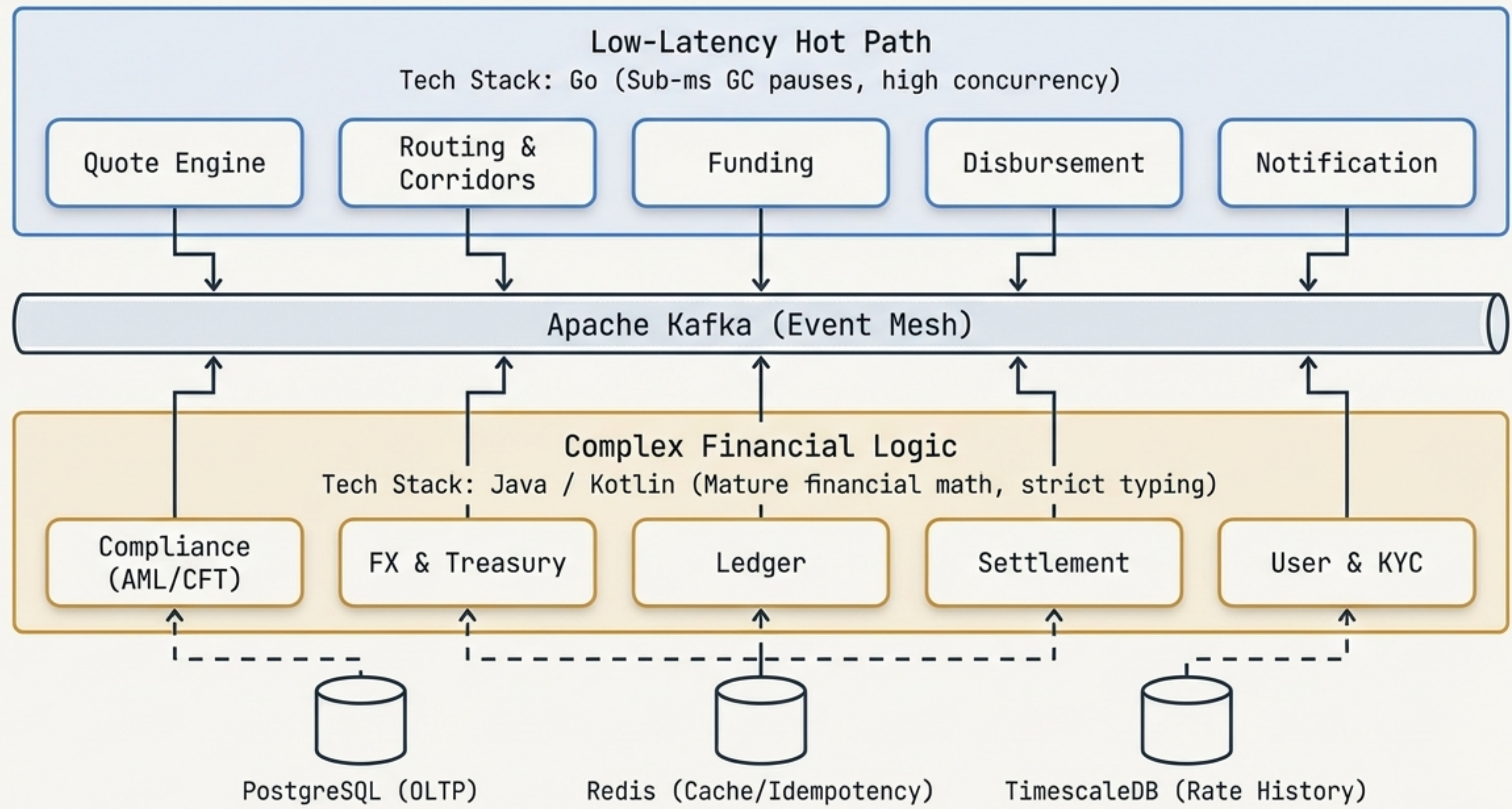
Latency Constraints

- Quote Creation:
< 200 ms 
(p99)
- Compliance Screening:
< 300 ms 
for 4 parallel checks
- Target Delivery:
< 1 hour 
for major corridors

Reliability & Precision

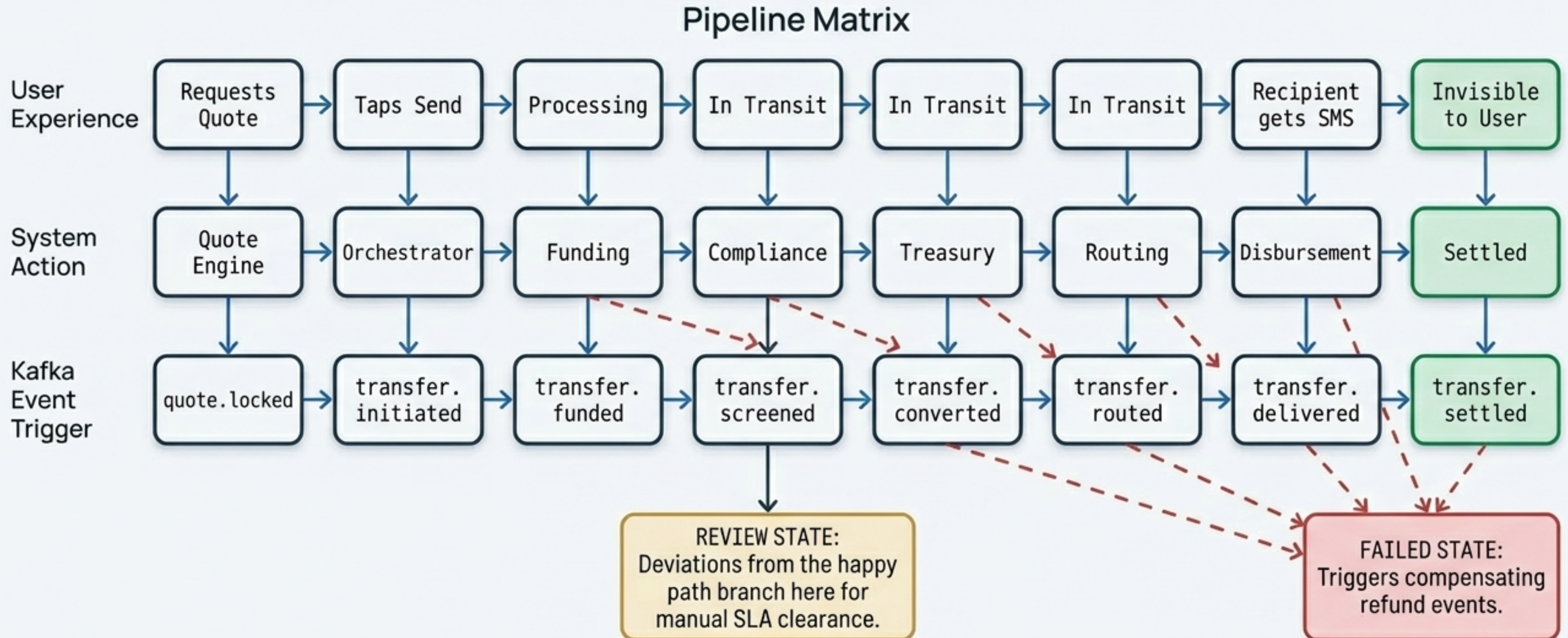
- Availability:
99.95%
uptime 
- Ledger:
Immutable
double-entry
bookkeeping 
- Audit Trail:
7-year WORM
cold storage
retention 

High-Level Topology: 10 Microservices Bound by an Event Mesh

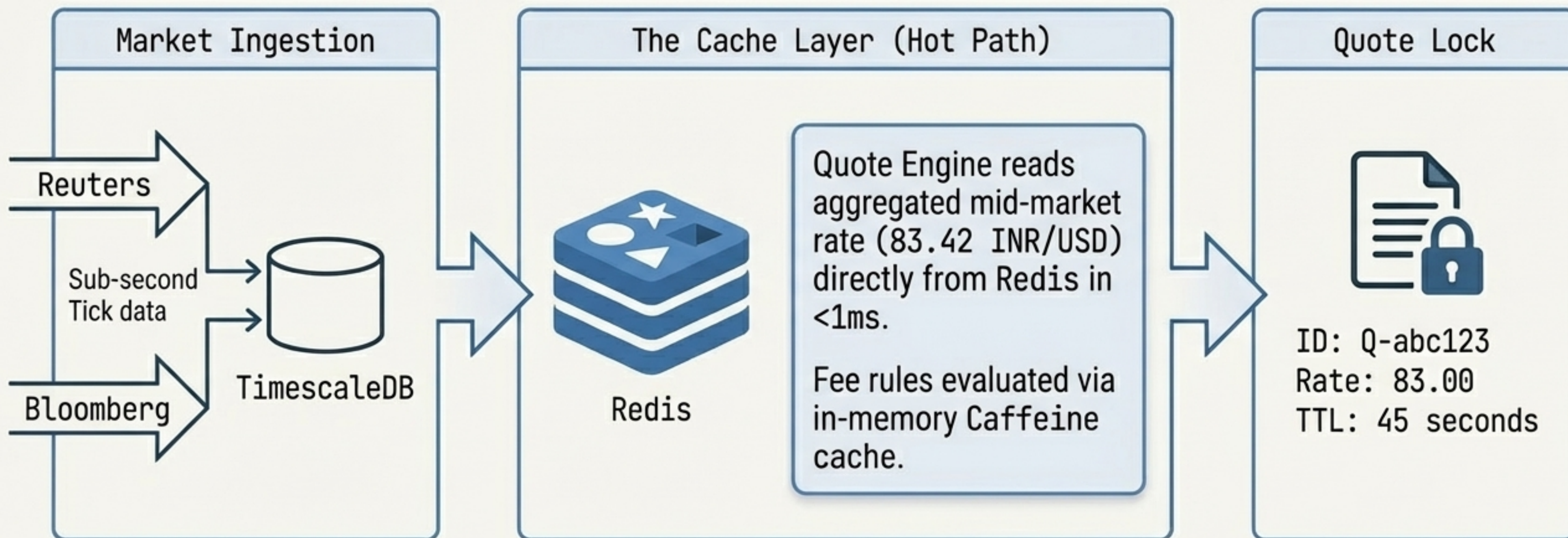


The Transfer State Machine: A \$1,000 Journey

Scenario: Priya (US) sends \$1,000 to her father Rajesh (India) via ACH.



Stages 1 & 2: Sub-200ms Quoting & Initiation

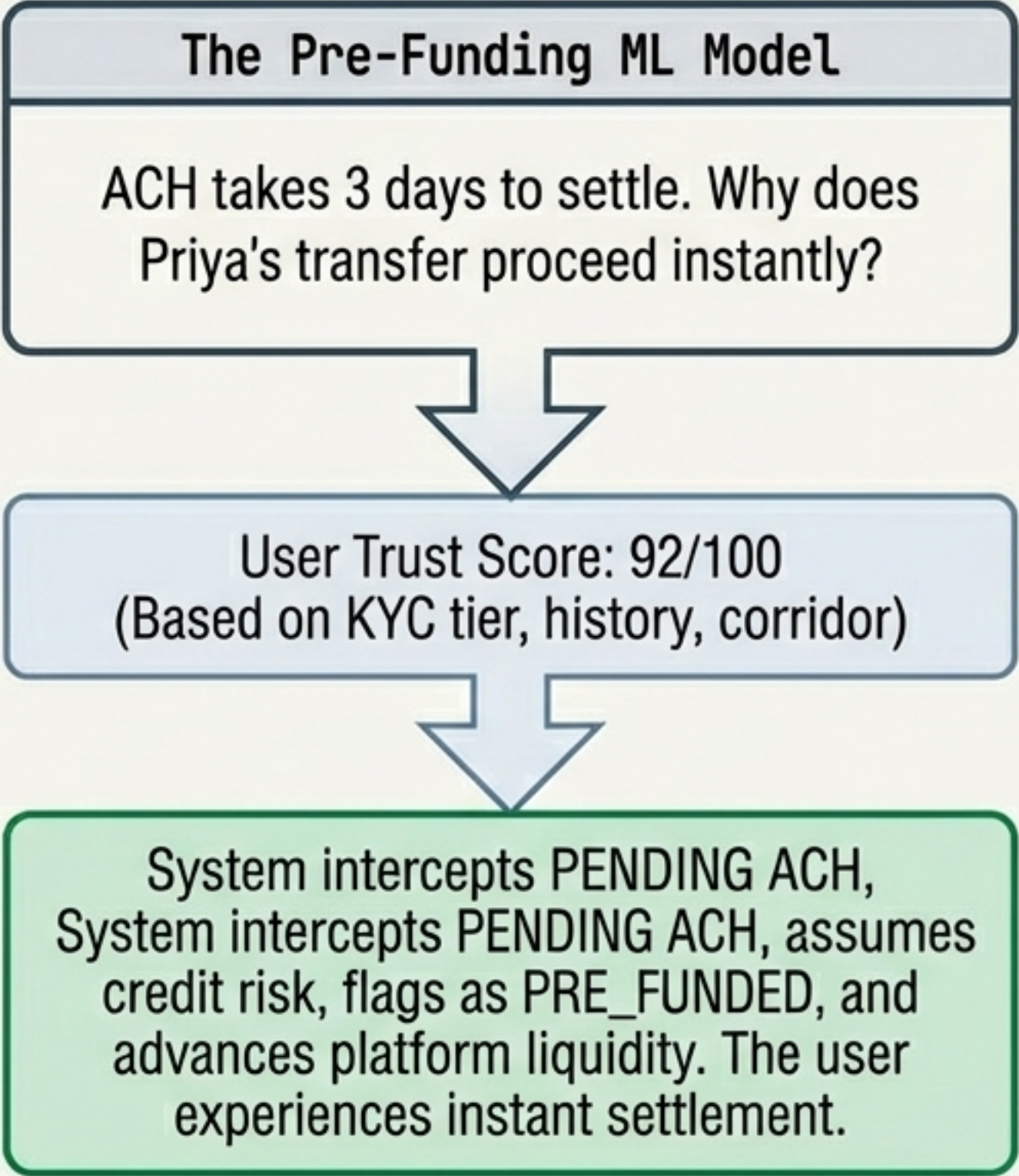


Insight: Why no database reads?

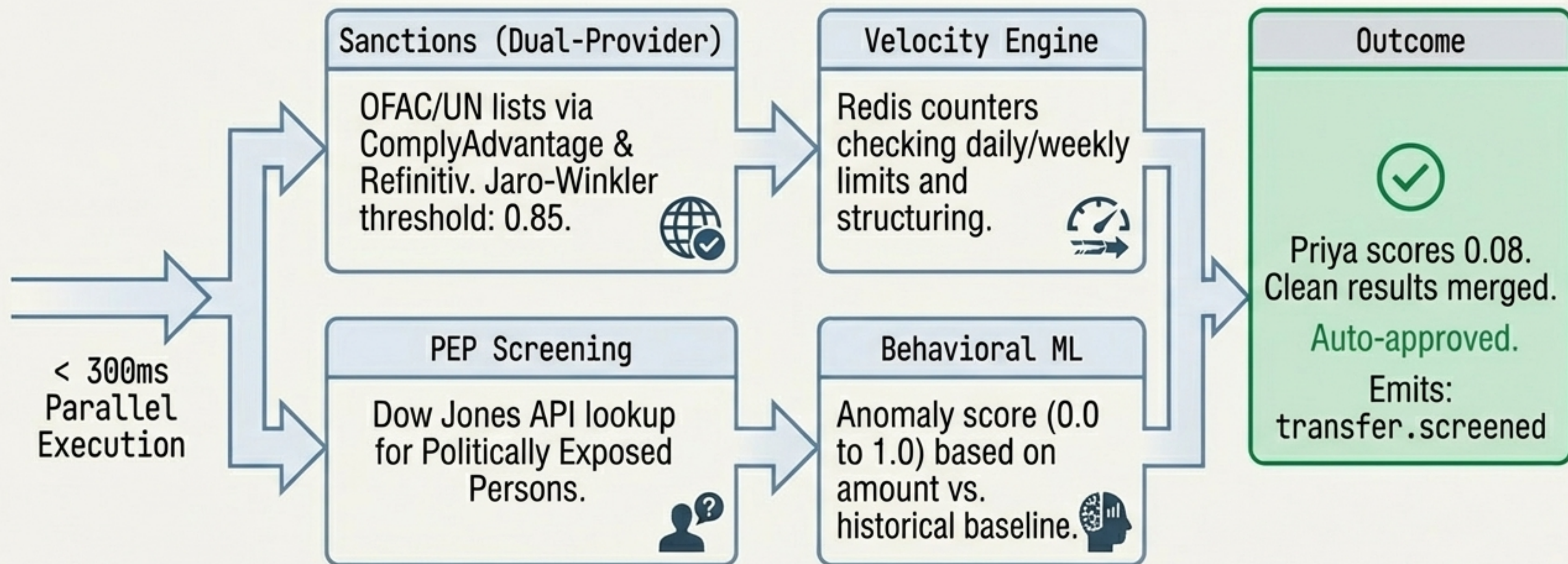
By keeping the hot path entirely in memory and Redis, the p99 latency is kept under **14ms**. The user sees an instant price. The transfer initiation locks this quote using an X-Idempotency-Key to prevent duplicate submissions.

Stage 3: Funding & Collection Economics

Diagnostic Comparison Matrix			
Method	Speed	Cost	Risk
Card (Stripe/Adyen)	Instant	High (~2.5%)	Chargebacks
ACH (Plaid)	1-3 Days	Low (~\$0.50)	NSF Returns (R01)
Bank Transfer	1-2 Days	Med (\$1-\$2)	Unmatched references
Open Banking	Instant	Low (£0.20)	Low

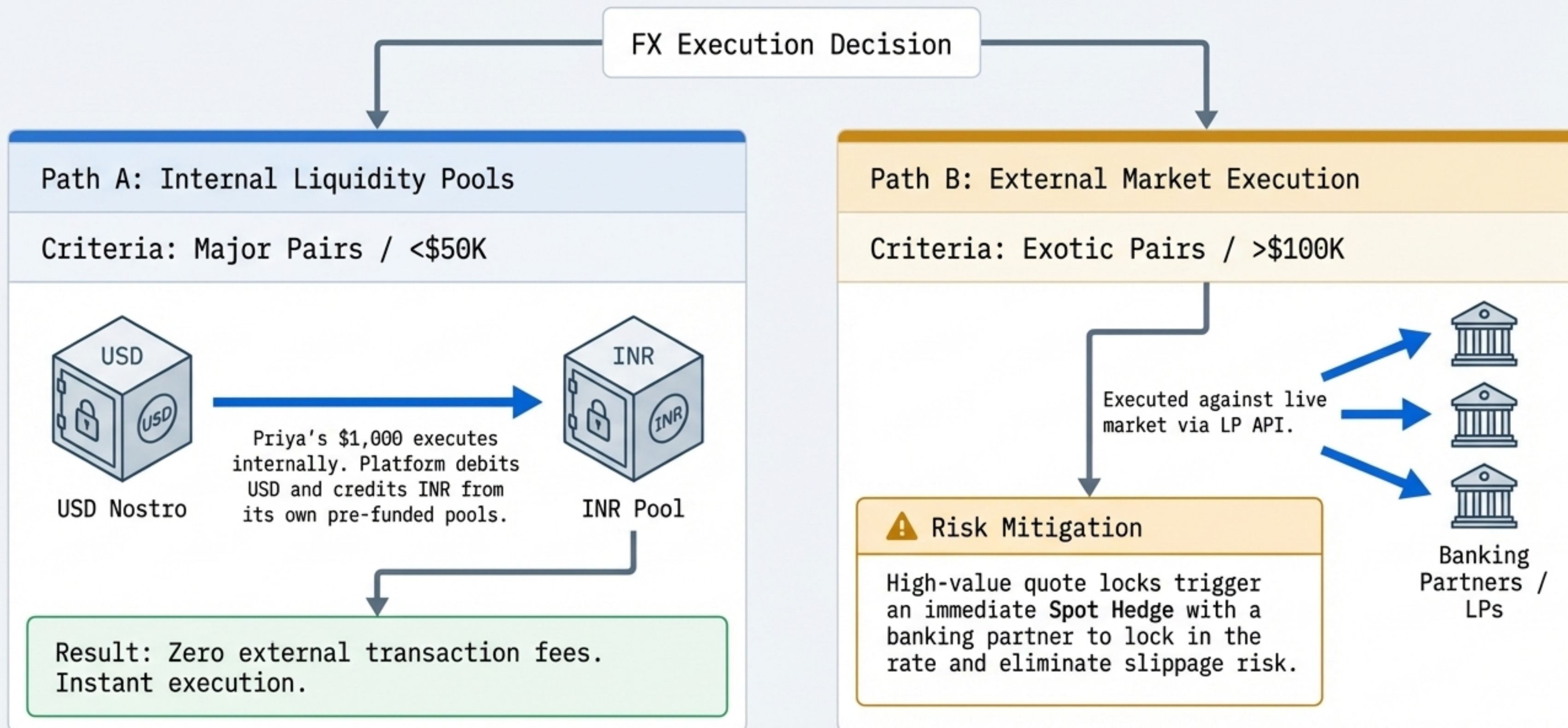


Stage 4: High-Speed AML & Compliance Engine



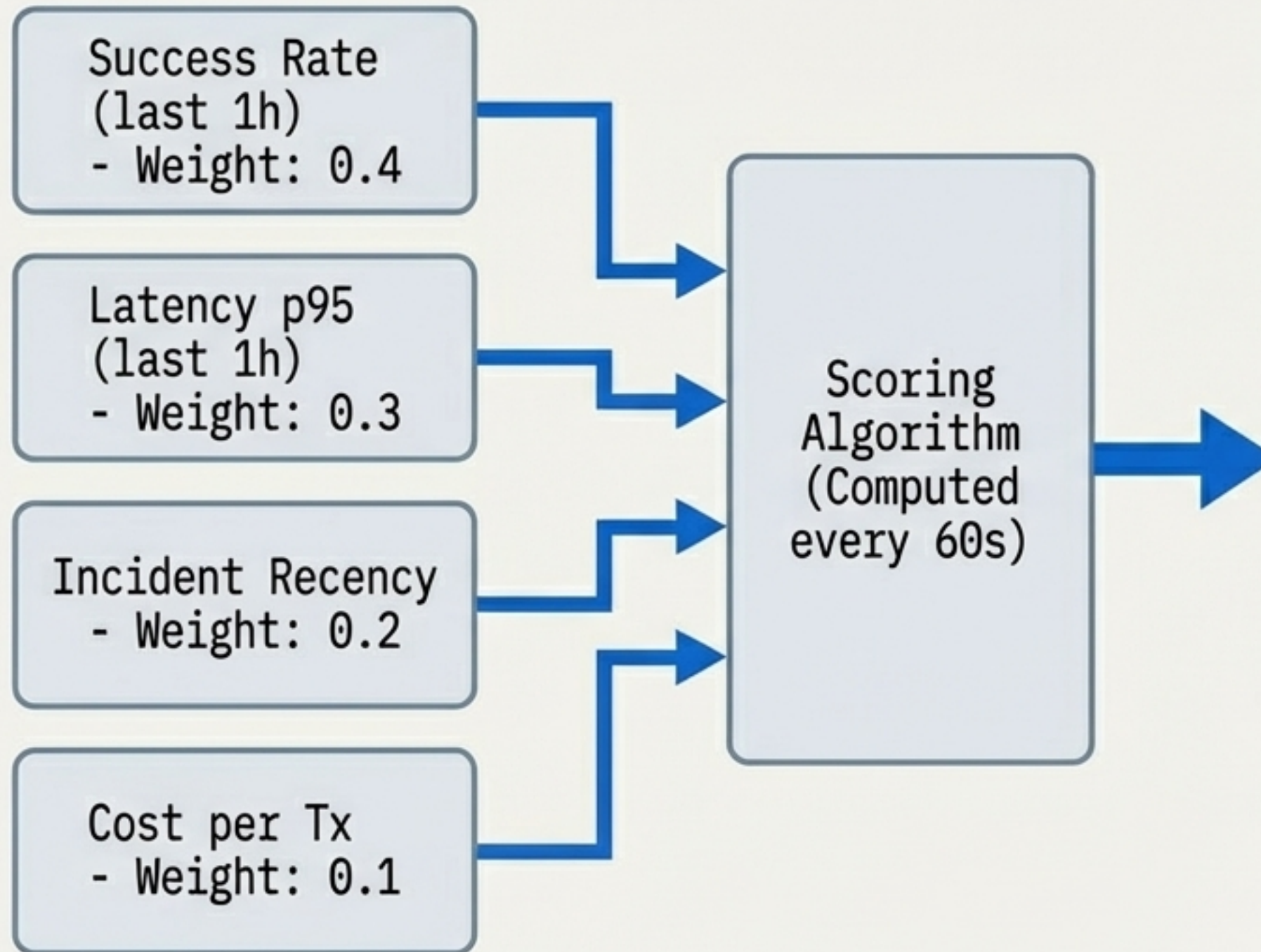
Failover Mechanism: If any provider times out or returns an anomaly score >0.7 , the transfer aborts the hot path and enters the REVIEW queue with a strict P1-P4 SLA for human analysts.

Stage 5: FX Execution & Liquidity Management



Stages 6-7: Smart Routing & Disbursement

Weighting Mechanism



Disbursement Execution

Ranked Failover List:

✓ **1. Winner:** Partner A (IMPS real-time rail) | Score: 0.95 | Selected for transfer.

2. Fallback: Partner B (NEFT batch rail) | Score: 0.88

⚠ Circuit Breaker Logic

If Partner A returns 5 consecutive failures, the circuit trips OPEN. The orchestrator seamlessly auto-fails over to Partner B, preventing corridor downtime.

Stage 8: Settlement & The Immutable Ledger

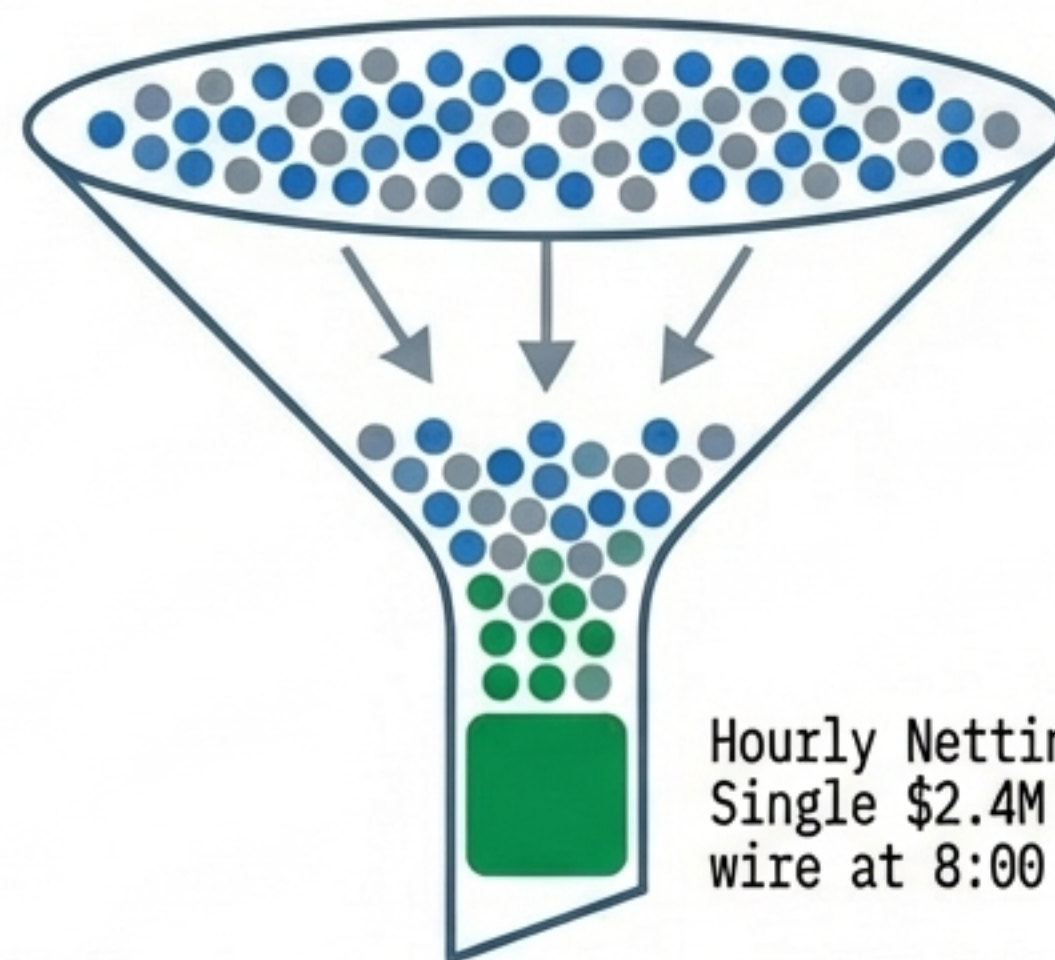
The Ledger (Double-Entry Invariant)

PostgreSQL Append-Only Ledger						
transaction_id	account	type	amount	currency	previous_entry_hash	timestamp
TXN_12345	USR_ACCT_A	DEBIT	100.00	USD	0xabc123...	2024-05-20 08:00:00 UTC
TXN_12345	SYSTEM_SETTLEMENT	CREDIT	100.00	USD	0xdef456...	2024-05-20 08:00:00 UTC
...	...	...	...	...	...	...
TXN_12345	USR_ACCT_B	DEBIT	100.00	USD	0xabc123...	2024-05-20 08:00:00 UTC
TXN_12345	SYSTEM_SETTLEMENT	CREDIT	100.00	USD	0xdef456...	2024-05-20 08:00:00 UTC
...	...	...	...	...	...	...

Every action creates offsetting debit/credit rows. Entries contain SHA-256 hashes of previous entries for tamper detection. UPDATE and DELETE commands are strictly blocked at the DS DB role level.

Batched Settlement Netting

3,200 individual daily cross-border wires



Hourly Netting ->
Single \$2.4M wholesale
wire at 8:00 PM UTC.

Takeaway: Netting drastically cuts bank wire fees, unlocking the unit economics.

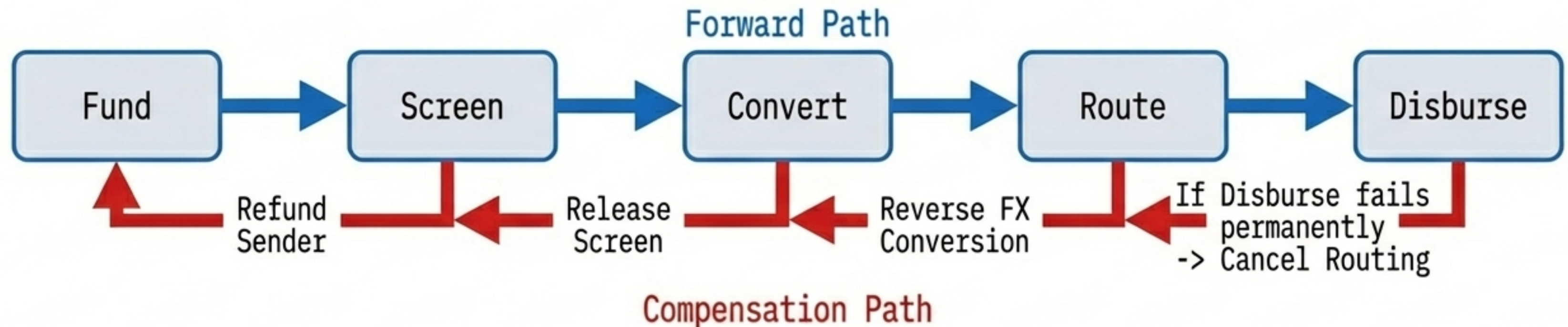
Fault Tolerance: The Orchestrator Saga Pattern

Why Not 2-Phase Commit (2PC)?

External partner APIs do not support XA transactions.

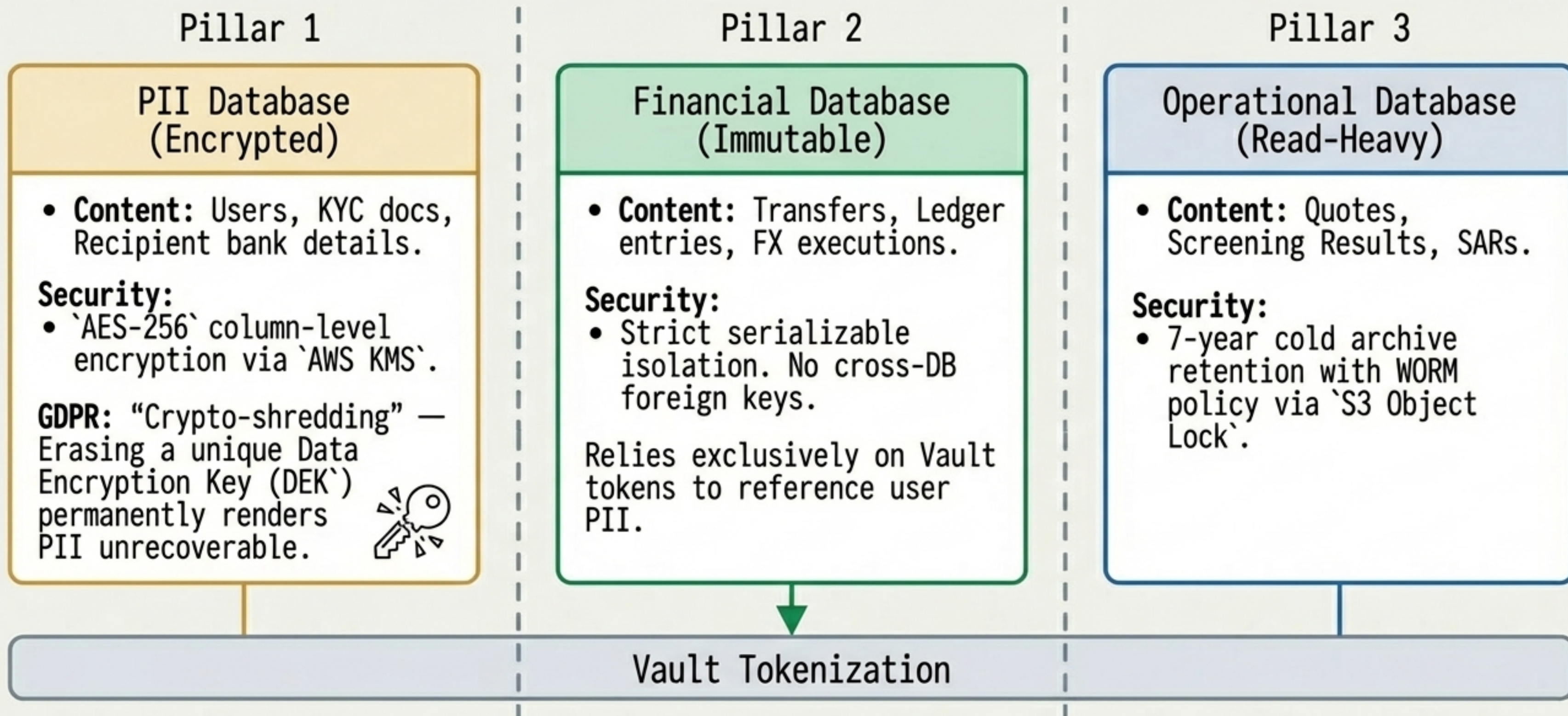
2PC would block entirely if a single emerging-market payout partner went down.

Event-Driven Choreography

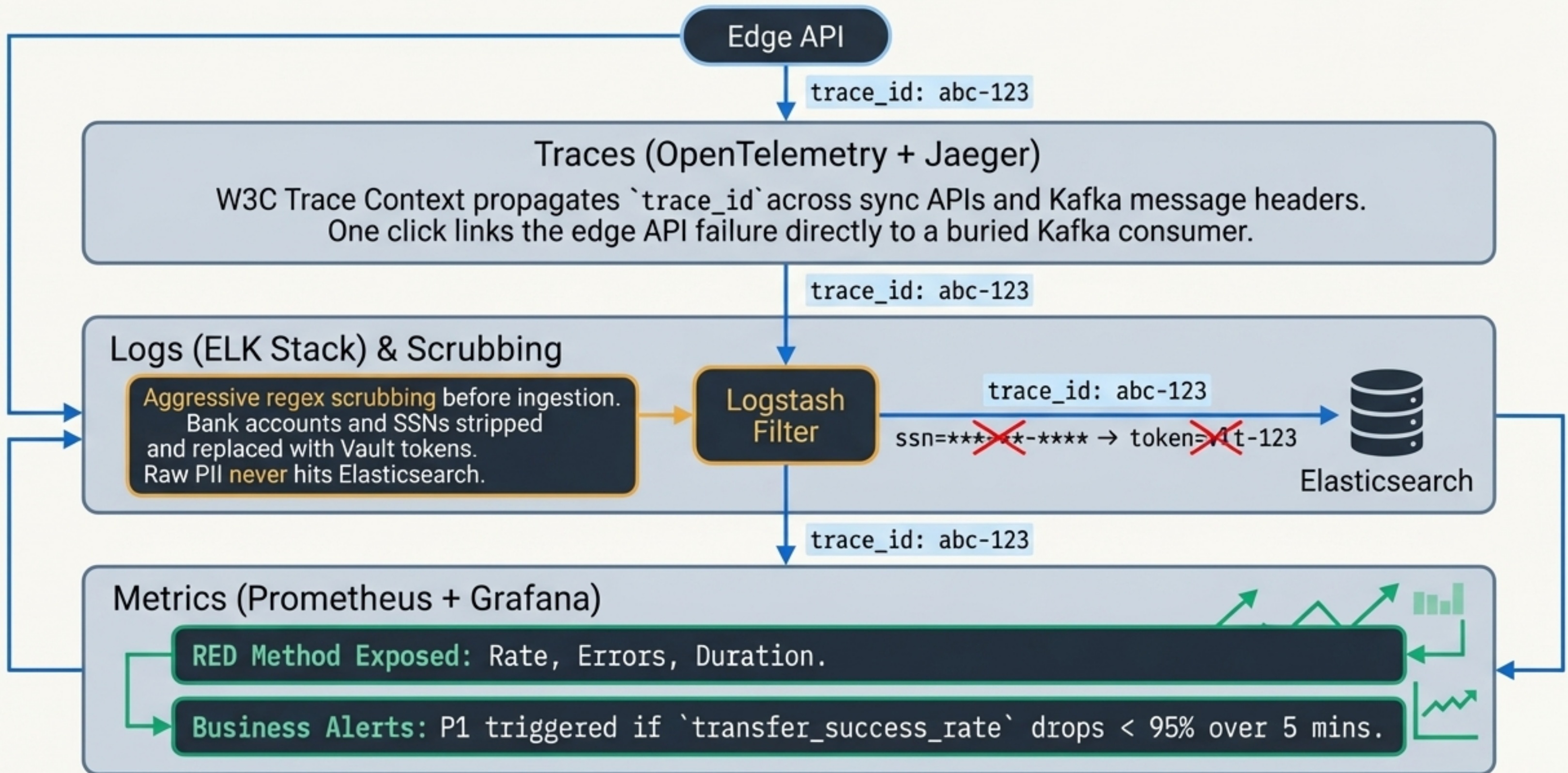


Architectural Note: Every node relies on a Redis-backed 24h X-Idempotency-Key to guarantee network retries never result in double-charging or double-payouts.

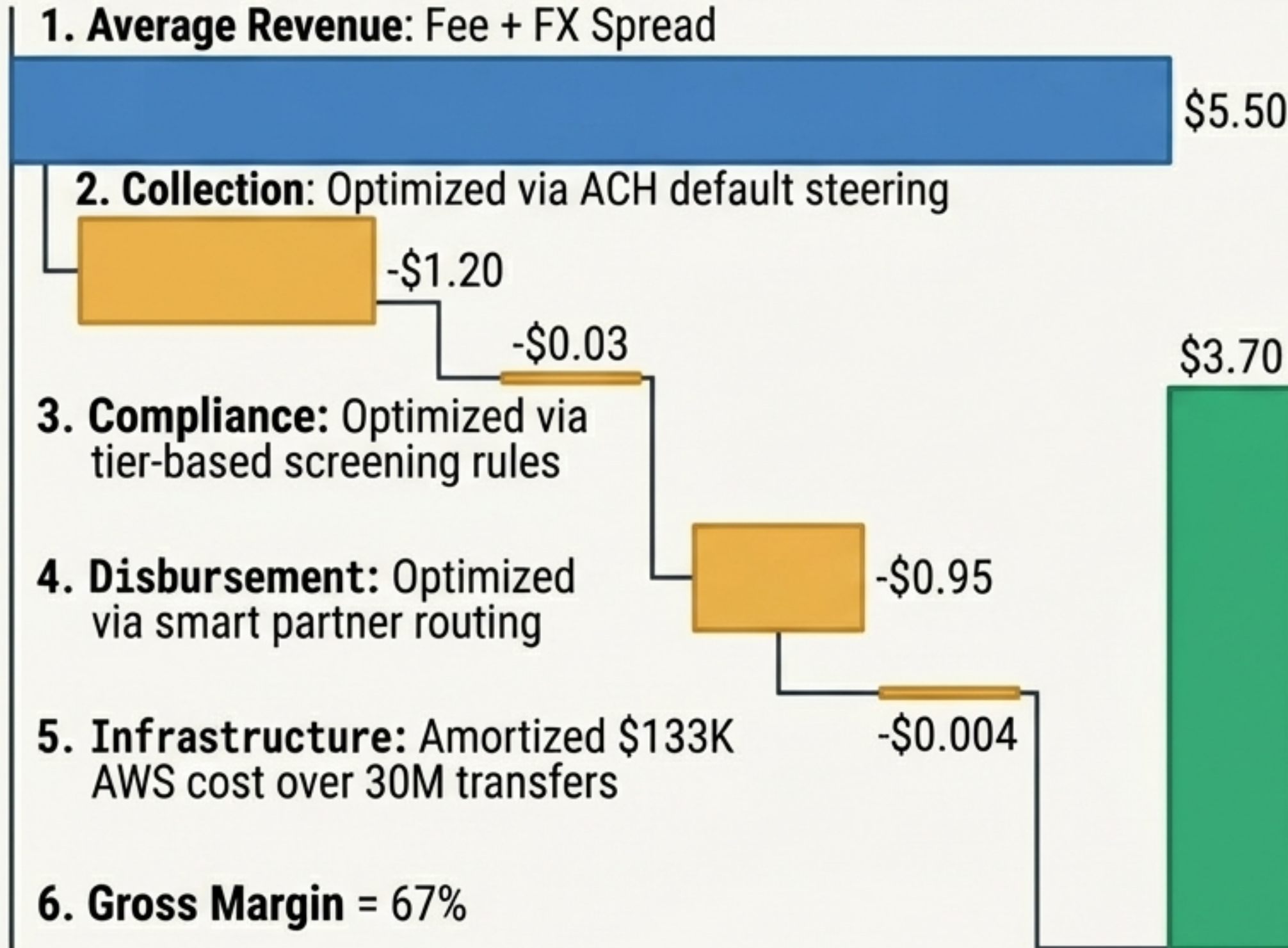
Data Isolation & Security Boundaries



Distributed Observability & PII Scrubbing



Unit Economics at Scale: Proving the Architecture

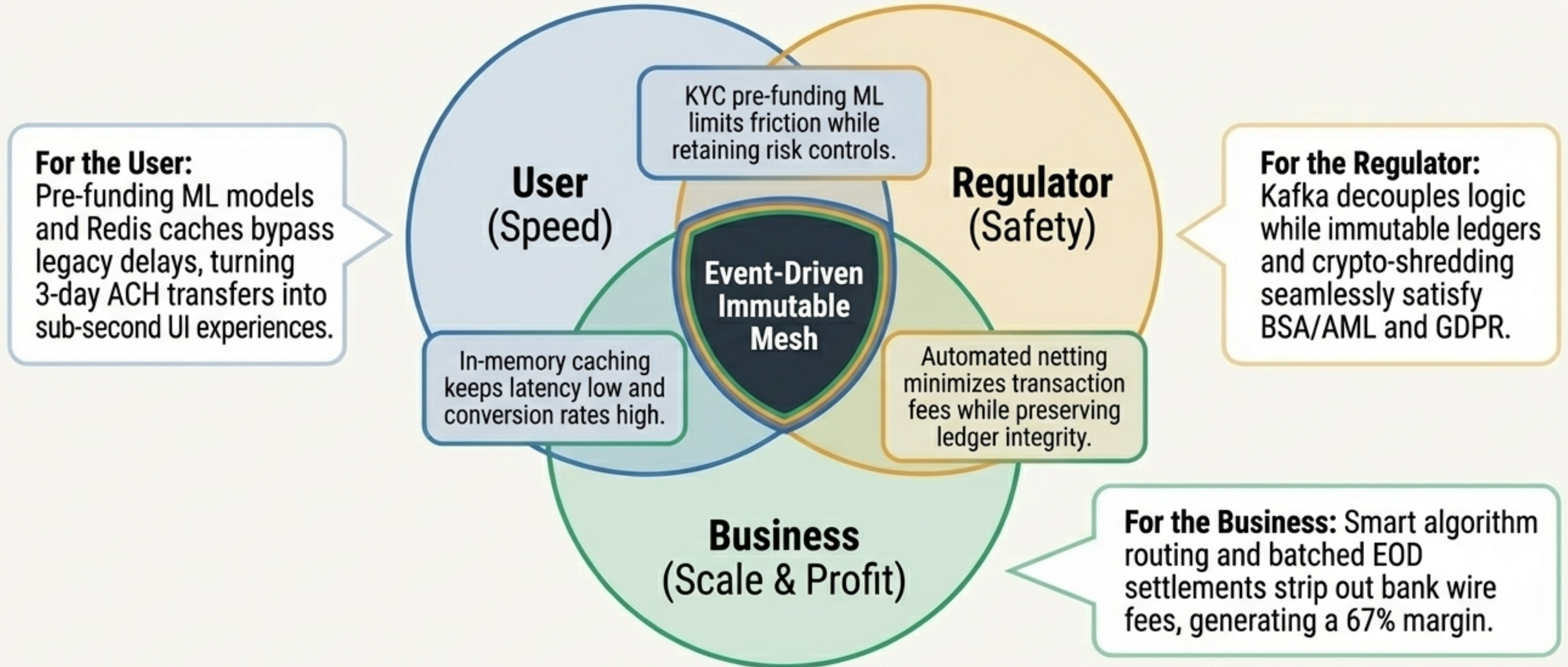


Architectural Leverage

Intelligent routing, batched settlements, and optimized compute scaling transform a complex global transaction into a highly profitable digital product.

Infrastructure cost per transfer is driven to near-zero (\$0.004) at scale.

Architectural Synthesis: The Fintech Balance



A system simultaneously fast for users, safe for regulators, and highly profitable for the business.